Aditya Mitra

Email | LinkedIn | GitHub

Education

KIIT University — Bhubaneswar, Odisha, India
B.Tech in Computer Science & Engineering — ***CGPA: 9.5/10*** — *Expected May 2029*

Projects

SentinelML | *XGBoost, SHAP, FastAPI, MLflow, Docker*
- Built a network intrusion detection system reaching **94.2% recall** at a low **1.5% false-positive rate** on CICIDS, using an XGBoost classifier with SHAP-based per-alert explanations for analyst triage.
- Served the model through a containerized **FastAPI** inference API (p95 latency **24 ms**) with **MLflow** experiment tracking and model registry for reproducible releases.

FraudLens | *Kafka, Scikit-Learn, Python*
- Designed a streaming financial anomaly detection pipeline on **Kafka**, improving **PR-AUC** from **0.68** to **0.87** on highly imbalanced data via threshold tuning against a business **cost matrix**.
- Evaluated models on PR-AUC and expected monetary loss instead of accuracy, reducing modeled fraud cost by **32%** versus the baseline.

PredictiveOps | *PyTorch, Feature Store, Prometheus, Python*
- Developed a time-series anomaly detection and predictive-maintenance system for IoT sensor data in **PyTorch**, detecting faults **18** hours ahead with **91%** precision.
- Added a feature store for train/serve consistency and **Prometheus** metrics for model and pipeline health monitoring.

CampusMind | *Qdrant, HuggingFace, FastAPI, OCR, RAG*
- Built a multimodal campus assistant combining **OCR** ingestion with a **hybrid RAG** agent (dense + keyword retrieval on **Qdrant**), reaching **89%** answer accuracy on a **250**-question evaluation set.
- Exposed retrieval and generation through a **FastAPI** service using HuggingFace models, with sub-**0.4**s median response time.

AutoML-Studio | *Optuna, Scikit-Learn, Streamlit, MLflow*
- Created a tabular AutoML orchestrator that automates preprocessing, model selection and **Optuna** hyperparameter search, cutting manual tuning time by **65%** across **12** benchmark datasets.
- Logged every run to **MLflow** and shipped a **Streamlit** UI so non-experts can train, compare and export models.

Student Marks Manager & Metadata Extractor | *Java, Spring Boot, MySQL, REST APIs*
- Implemented layered **Spring Boot** services with **14** RESTful endpoints and normalized **MySQL** schemas for student records, marks and metadata extraction.
- Added input validation and error handling across the API, and covered core service logic with **28** unit tests.

Experience & Open Source

System Developer — Sept 2026 – Present
GeeksforGeeks (GFG) Student Chapter, KIIT — *Bhubaneswar, India*
- Built and maintained **4** chapter internal automation tools and event workflows used by **350+** active members.

Open Source Contributor — Aug 2026 – Present
GirlScript Summer of Code (GSSoC) & Apache Ecosystem — *Remote*
- Merged **8** pull requests across open-source repositories, focusing on documentation improvements, unit testing, and bug fixes.

Competitive Programmer — Oct 2026 – Present
Coding Ninjas, KIIT Chapter — *Bhubaneswar, India*
- Solved **200+** algorithm problems across LeetCode and platforms (focusing on arrays, graphs, and dynamic programming).

General Volunteer — Aug 2026 – Present
National Service Scheme (NSS), KIIT — *Bhubaneswar, India*

Certifications & Courses

Certified Kubernetes Administrator (CKA) – KodeKloud, Feb 2026 • **AWS Certified AI Practitioner** – KodeKloud, Feb 2026
Claude Code in Action – Anthropic, Apr 2026 • **AI Fundamentals** – IBM SkillsBuild, Apr 2026
Introduction to Data Science & IoT Analytics – Infosys Springboard, Apr 2026

Technical Skills

Languages: Java, Python, SQL, C/C++
Backend: Spring Boot, FastAPI, REST APIs, MySQL, Kafka, Qdrant
DevOps / MLOps: Docker, Kubernetes, MLflow, Prometheus, Feature Stores, Git, Linux, CI/CD
ML / AI: PyTorch, Scikit-Learn, XGBoost, SHAP, Optuna, HuggingFace, RAG, OCR
Cloud: AWS (AI/ML fundamentals), Kubernetes cluster administration